DESIGN REPORT Concept – Sky

Sky is selected for XR redesign due to its natural alignment with spatial interaction paradigms. Socially, unlike traditional games reliant on text boxes, Sky emphasizes proximity-based non-verbal communication via body language, which seamlessly matches VR's native spatial logic. In terms of gameplay, its core mechanics—open-sky exploration, wind flow navigation, and natural environmental interactions—reach their full potential when controlled through 3D spatial hand gestures.

The redesign of Sky in XR focuses on three primary user tasks and goals. **First,** for Open-World 3D Exploration and Navigation, the goal is to provide a highly immersive, unrestricted aerial flight experience while minimizing motion sickness; users adjust viewing angles via the Head-Mounted Display (HMD), perform vertical takeoffs and obstacle traversal with Dual 6DoF Motion Controllers, and seamlessly transition between hovering and gliding through natural physical gestures. **Second,** for Embodied Non-Verbal Social Connection and Environmental Customization, the goal is to cultivate deep emotional connections with players and NPCs by replacing 2D menus with intuitive 3D spatial interactions; users perform proximity energy-sharing rituals ,such as heart-light docking, to initiate friendships, respond to spatial prompts with physical gestures to unlock NPC narrative trees, and use controllers to directly place social props like tea tables, into the 3D environment. **Third,** for Multi-modal In-Game Communication and Text Input, the goal is to deliver efficient, non-intrusive real-time dialogue without disrupting visual immersion; users communicate hands-free via Automatic Speech Recognition during active gameplay and utilize discrete text-input methods in quiet environments to avoid obtrusive 2D popup keyboards.

The development of this XR concept progressed through a clear, iterative refinement process to transition the initial idea into a fully realized redesign for Sky. Initially, inspired by the spatial mechanics in Half-Life: Alyx, I identified diegetic interaction patterns suitable for Sky's environment. In the next iteration, I evaluated VR versus MR, selecting a pure VR environment to maintain the game's artistic immersion without physical real-world spatial constraints. I then mapped these core mechanics to specific XR hardware capabilities, combining a Head-Mounted Display, Dual 6DoF Motion Controllers, and a Finger-worn Motion Controller. Through visual concept development, I created sketches overlaid on 2D screenshots to translate traditional HUD elements into intuitive 3D spatial gestures. To validate these interactions, I built low-fidelity paper prototypes to physically simulate spatial mechanics such as perspective switching. Finally, user testing revealed that relying exclusively on voice input excluded speech-impaired players and restricted quiet environments; thus, I refined the concept into a multi-modal input system that incorporates a finger-worn controller for real-time air-typing alongside speech recognition.

This XR redesign translates Sky into a pure Virtual Reality environment to maximize emotional and spatial immersion. Paired with a designated play boundary, VR

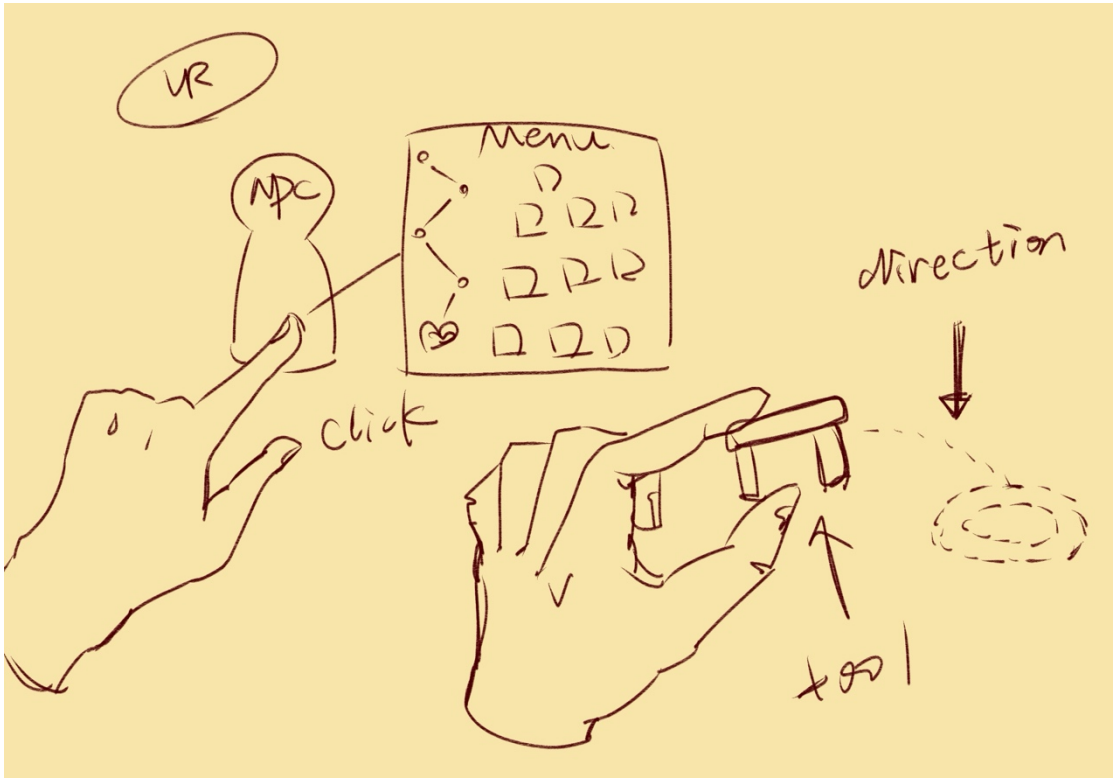
eliminates safety concerns about real-world obstacles, enabling players to freely perform large-scale physical gestures like spreading arms to glide or reaching out to share heart-lights. Furthermore, because Sky's core interactions—gliding in wind currents, lighting candles, and collecting floating energy—rely entirely on virtual physical laws, interacting with real-world objects is unnecessary.

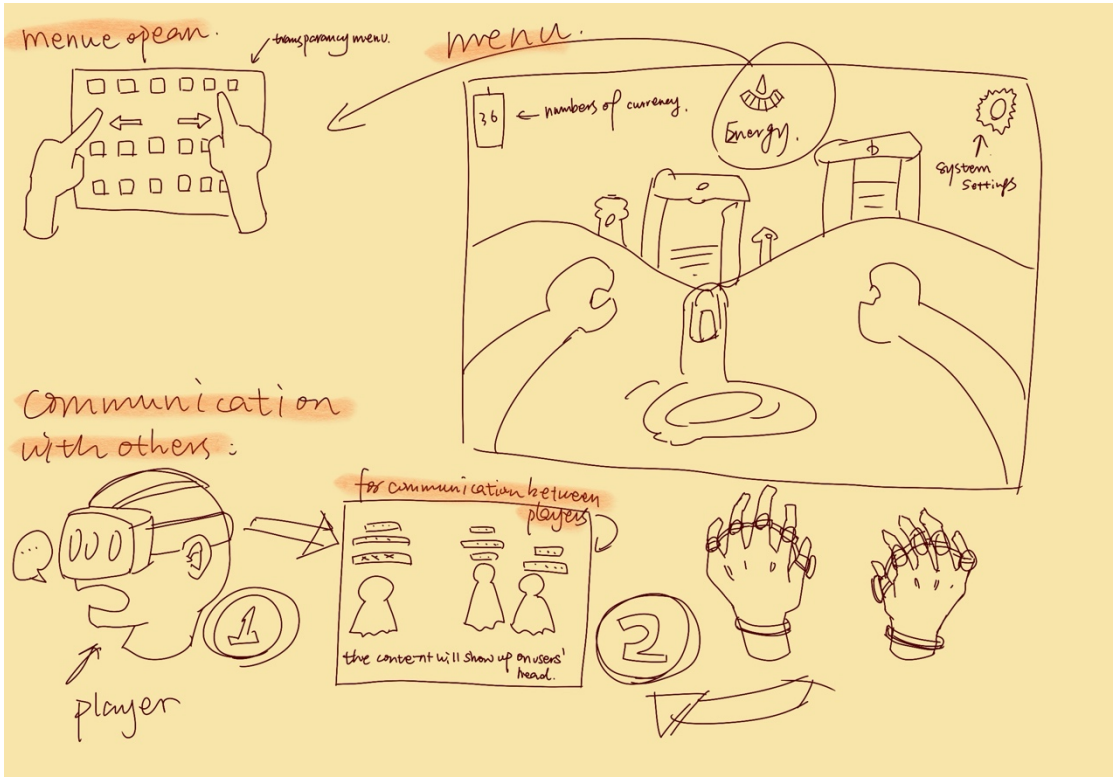
To support these user tasks, the VR environment utilizes tailored spatial interactions and physical affordances. For navigation and flight, users hold the 6DoF controller triggers and fling their hands downward to launch vertically, spreading their arms to steer gliding trajectories via controller pitch and roll, while haptic feedback signals wind resistance and dynamic vignetting reduces motion sickness during high-speed dives. For embodied social interaction, users bring a controller near their chest to initiate proximity "heart-light" docking, extend hands forward to offer white candles for friend requests, use pinch gestures or grip buttons to manipulate 3D spatial menus for outfit changes or placing props like tables, and perform physical gestures such as waving, guided by floating glowing auras to interact with ancestral NPCs. Finally, for communication, users utilize wearable finger-worn controllers to enable discrete air-typing text input alongside real-time voice conversion.

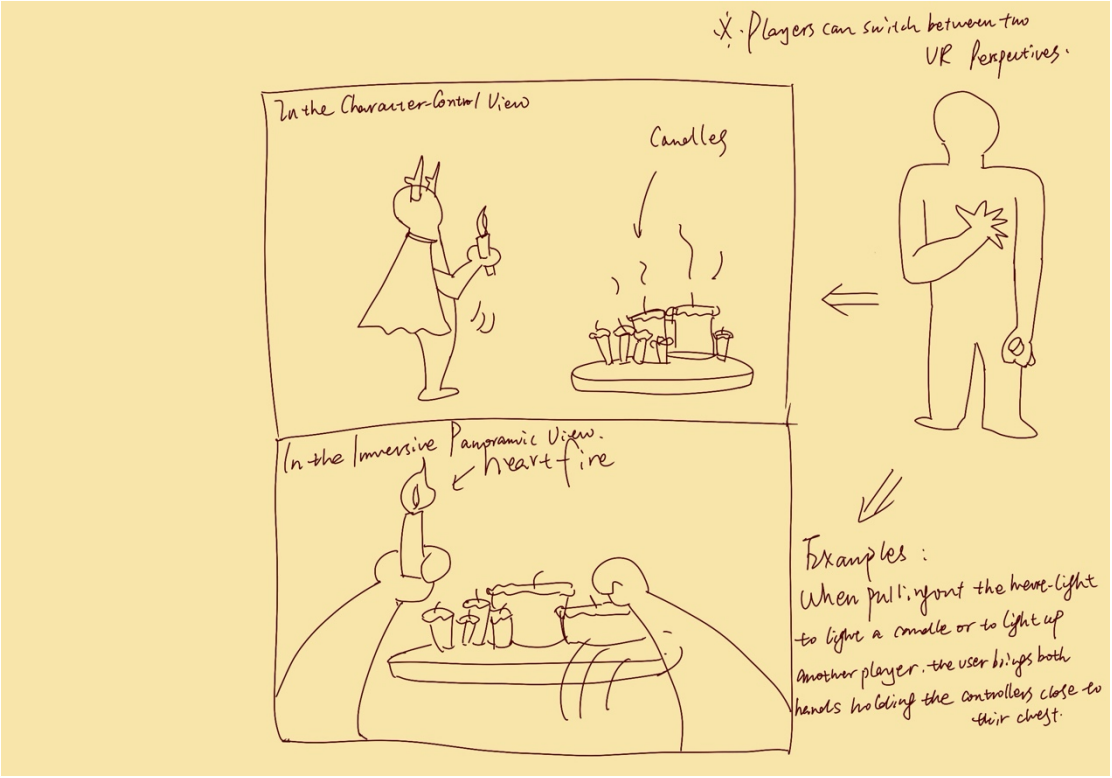
The specific features to test include validating the transition smoothness, spatial orientation, and visual comfort of the dual-perspective switching mechanism between Third Person Follow View and First-Person Embodied View. Additionally, the testing will evaluate the multi-modal communication system by comparing the usability and text-entry efficiency of finger-worn air-typing against Automatic Speech Recognition across both quiet and noisy settings. Finally, it will assess the gesture sensitivity, responsiveness, and spatial accuracy when summoning the 3D mid-air spatial menu via overhead gestures and interacting with ancestor item trees. This concept assumes that dual-perspective switching balances spatial orientation with visual comfort and social immersion; finger-worn air-typing provides a reliable text-entry alternative when ASR is hindered by noise or quiet constraints; and overhead gestures for 3D spatial menus deliver an intuitive, non-intrusive interaction flow. Validating these assumptions requires testing whether perspective transitions induce motion sickness, comparing air-typing efficiency against ASR, and assessing overhead gesture latency and fatigue. Data collected will include quantitative metrics—Task Completion Time, text-entry error rates and WPM, gesture latency, and Simulator Sickness Questionnaire scores—alongside qualitative feedback on spatial awareness, menu responsiveness, and arm comfort.

APPENDIX

Sketches of initial design

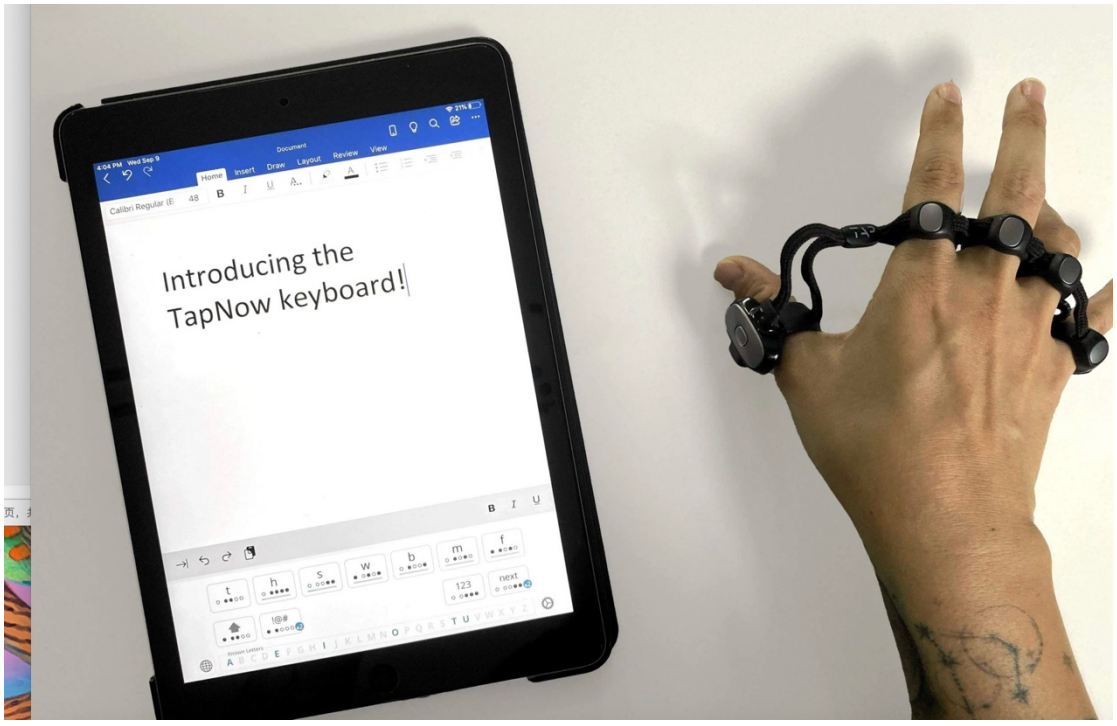

VR
Menu
NPC
direction
click
tool

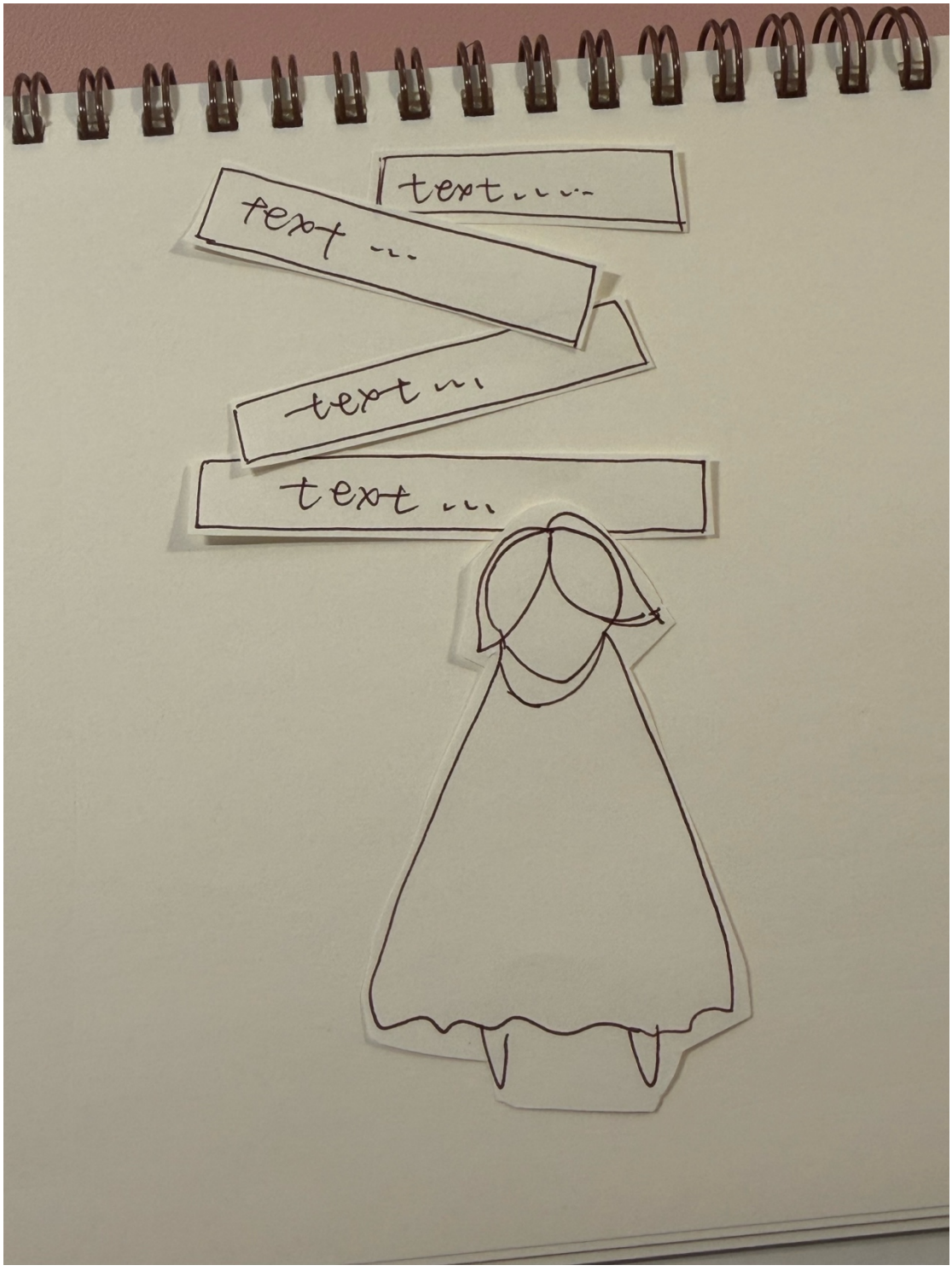

menue opean.
transparency menu.
menu.
36 ← numbers of currency.
Energy.
system settings
Communication with others:
for communication between players
the content will show up on users' head.
player
1
2



Inspirations from Finger-worn Motion Controller



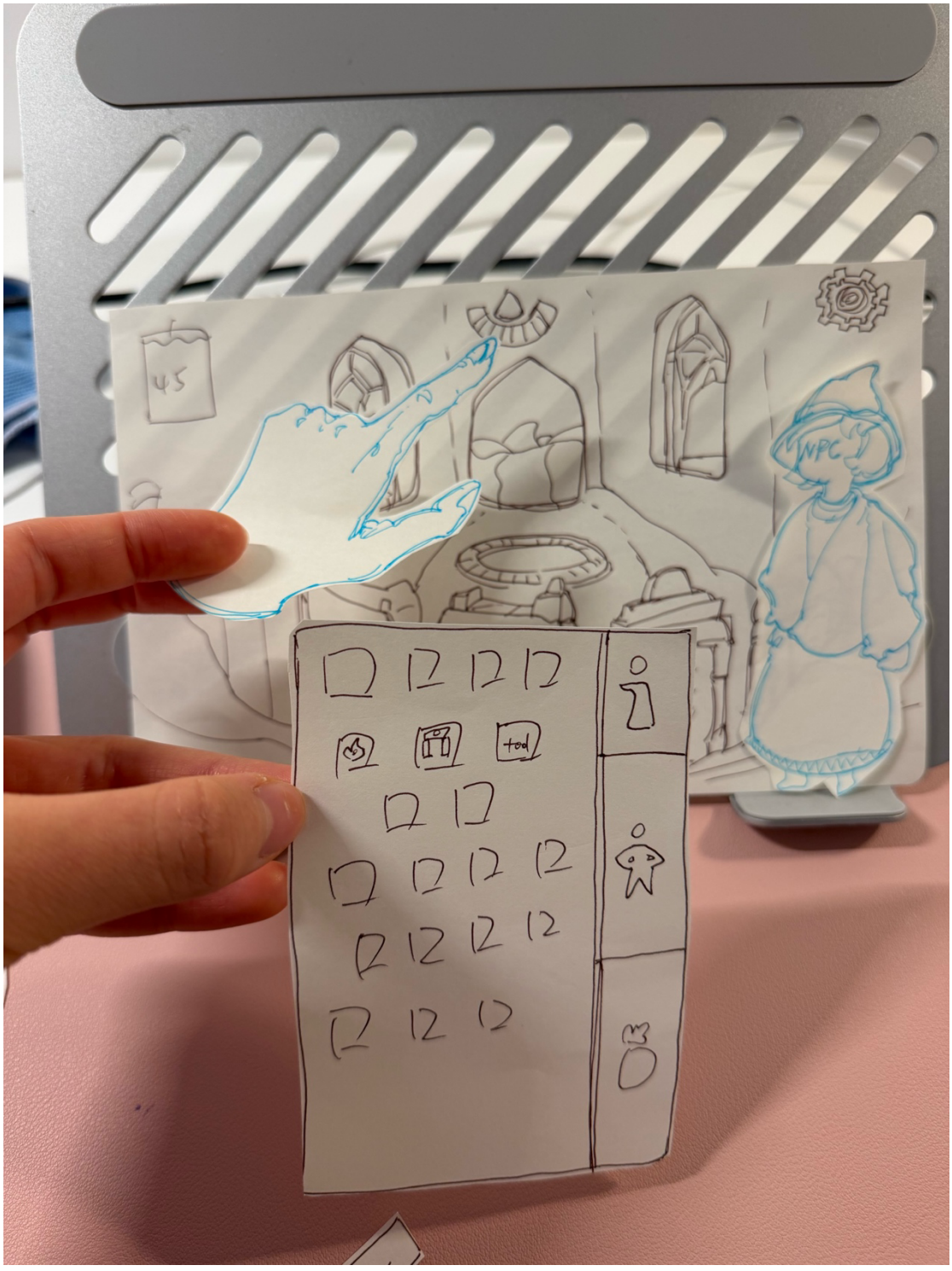

Introducing the
TapNow keyboard!



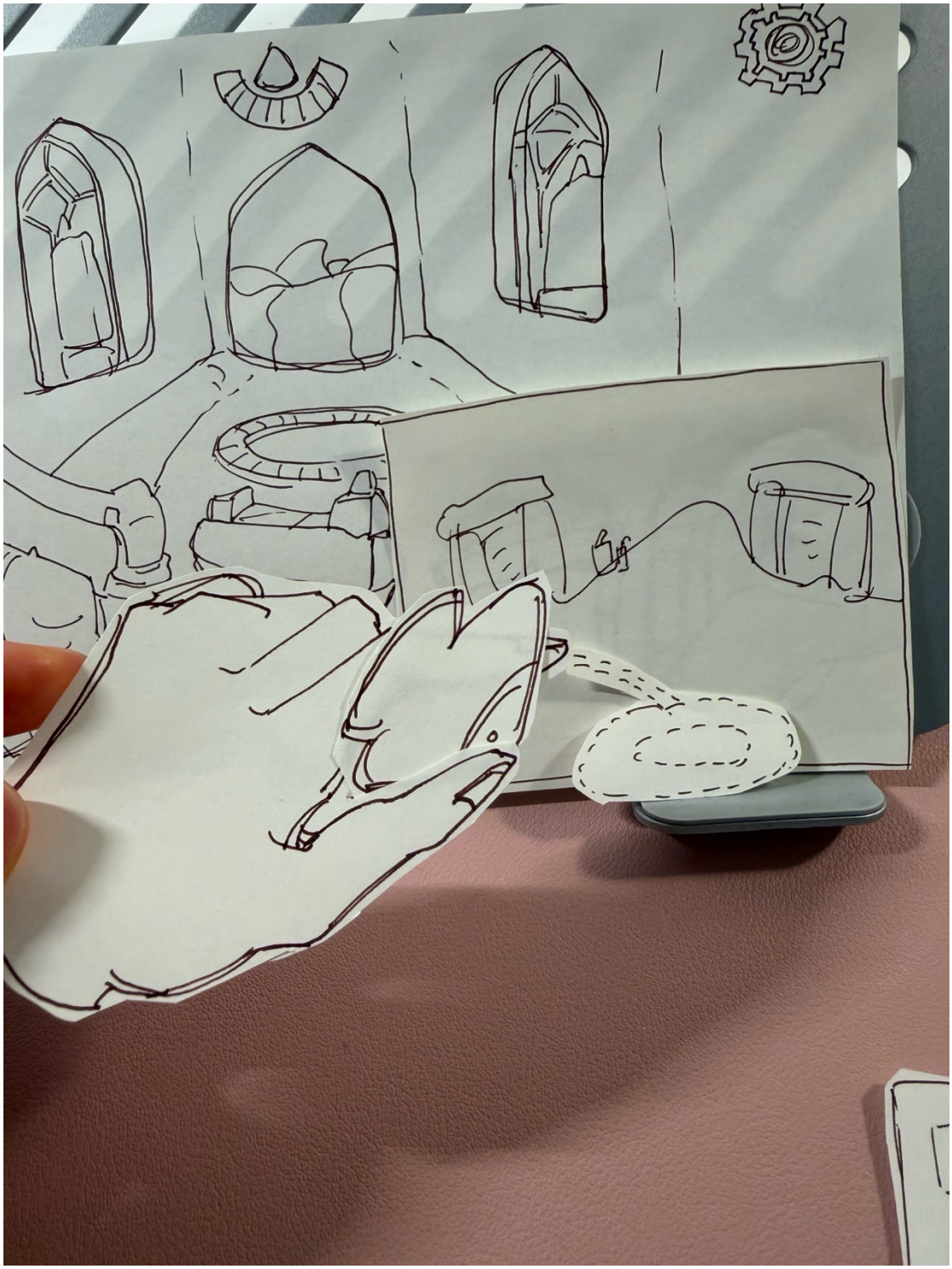
Low-fidelity prototyping of in-VR real-time translation



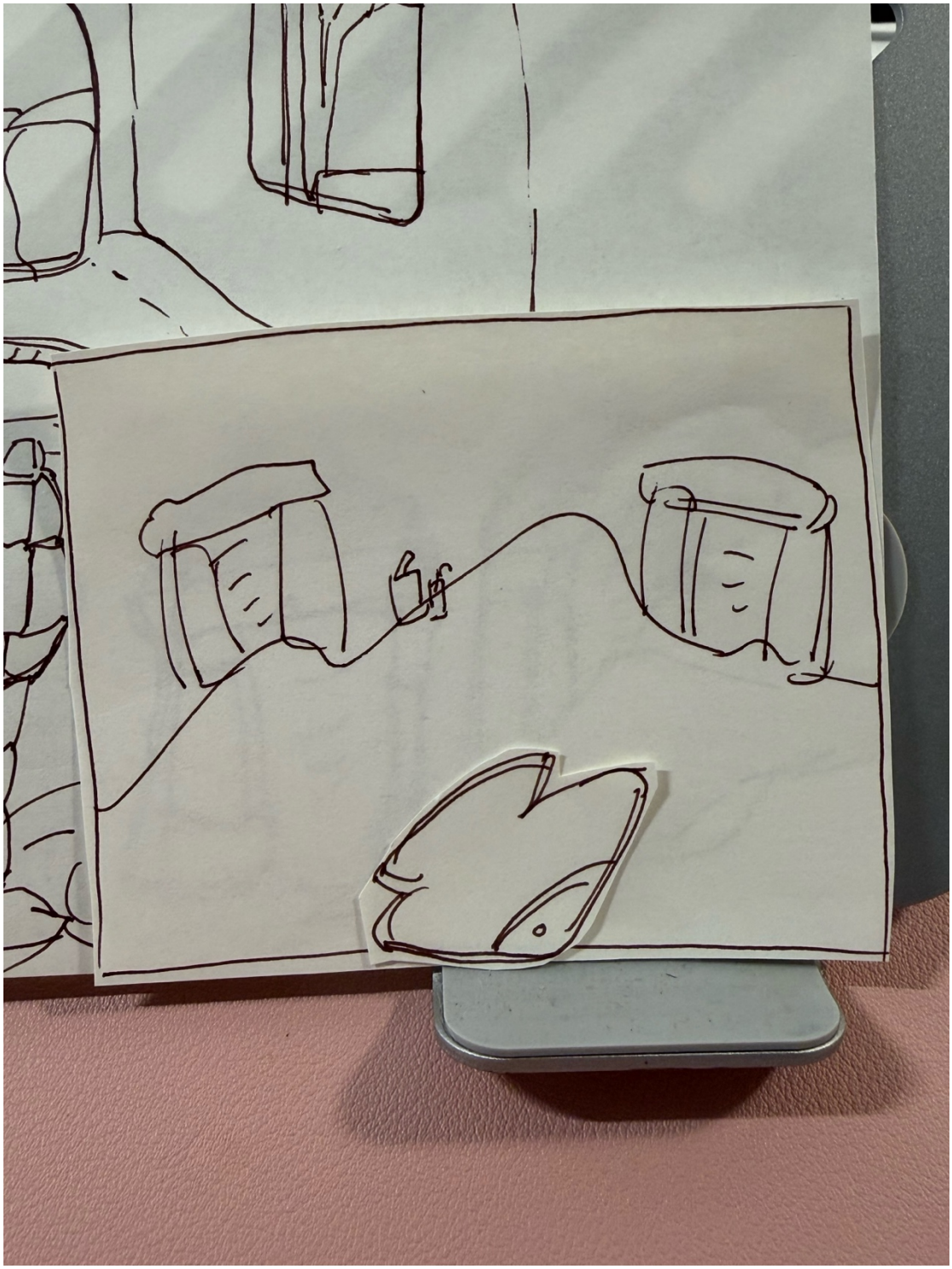
Low-fidelity simulation of immersive NPC interaction to summon the node tree



Low-fidelity simulation of summoning the VR UI top menu in immersive view



Low-fidelity simulation of prop placement from an immersive perspective1



Low-fidelity simulation of prop placement from an immersive perspective2